

KRISHNA AGARWAL

BCA UNDERGRADUATE

PROFESSIONAL SUMMARY

BCA undergraduate and Java backend developer with hands-on experience in building scalable web applications. Skilled in Java and Spring Boot with practical exposure to real-world projects.

CONTACT

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EDUCATION

National Post Graduate College 2023 - 2026	St. Paul's College 2009 - 2023	
Bachelor of Computer Applications	ICSE	ISC
CGPA : 8.2	95.1%	91.2%

PROJECTS

Shiksha Swap (In Progress)

Tech Stack: Java, Spring Boot, React.js, Tailwind CSS, Python, MySQL, AWS, Razorpay

- An educational marketplace with a custom matching engine that connects students with tutors for on-demand academic assistance.
- Implemented dual-role user system (Student + Tutor) with booking workflows

Agrosmart

Tech Stack: Java, Spring Boot, TypeScript, Tailwind CSS, Python, C++, IoT, MySQL

- An intelligent agriculture management platform designed to assist farmers with real-time monitoring, data-driven insights, and smart decision-making using IoT and web technologies.
- Integrated IoT-based data collection for real-time farm monitoring
- Integration of disease prediction and crop suitability models

NPGC College Website (Live Project)

Tech Stack: HTML, CSS, JS, Bootstrap, Python

- A responsive and user-centric institutional website designed for efficient management of academic information, student services, and campus updates for a modern college environment.
- Integration of NPFC Assistant - A Python RAG based Chatbot

Exampro - Online Examination System

Tech Stack: Java EE, Servlets, JSP, JSTL, MySQL, HTML, CSS, JS

- A full-stack enterprise-grade online examination and analytics platform featuring dual portals for teachers and students, real-time assessment engine, and anti-cheat proctoring system
- Implemented real-time exam engine with auto-save and smart countdown timer
- Built anti-cheat system using tab-switch detection and auto submission logic

Hospital Management System

Tech Stack: Java 17, JDBC, MySQL

- A console application simulating real-world hospital operations, including patient management, billing automation, and secure role-based workflows.
- Implemented Role-Based Access Control (RBAC) for Admin, Doctor, Nurse and Receptionist
- Designed layered architecture inspired by Spring Boot principles for modularity

Smart Waste Management

Tech Stack: MERN Stack, Arduino, IOT

- Developed an IoT-powered smart bin that detects human presence and automatically opens, reducing manual contact and contamination risks.
- Implemented real-time waste level monitoring using IoT sensors and displayed data on a web dashboard.
- Integrated Chart.js to provide a 7-day waste trend analysis

TECHNICAL SKILLS

- Programming - Java, JavaScript, SQL
- Frameworks - Spring Boot, Hibernate, Bootstrap, Node.js, Express.js, React.js
- Tools - Docker, Git/GitHub
- Web - REST APIs
- Architecture- MVC, Microservices

CERTIFICATIONS

DELTA - Full Stack Web Development

Apna College

- Intensive MERN Stack course covering frontend, backend, databases and deployment.

Java Fundamentals

Udemy - Abdul Bari

- Completed course covering object-oriented programming and core Java concepts

Java Spring Framework

Udemy - Navin Reddy (Telusko)

- Learned the fundamentals of Java , Advanced Java, Spring and Spring Boot and Backend Development

Java Full Stack

Eduskills Virtual Internship

- Learned the Java Full Stack Development with Java backend frameworks

Java Backend Development

Hanumant Technology Pvt. Ltd.

- Training Program of the backend development in Java and Spring Boot

RESEARCH AND EXPERIENCE

A Framework for Generalized Risk-Aware Planning and Decision-Making under Dynamic Adversarial Conditions

Annals of Computer Science and Intelligence Systems (Volume 14, Issue 03) - Scopus Indexed

- Proposed ARA++, a novel framework extending ARA* by integrating probabilistic enemy risk factor modeling and adaptive thresholding.

A Hybrid ElGamal – Paillier Approach for Achieving Practical Fully Homomorphic Encryption in Cloud Computing

Studies in Science of Science (Volume 43, Issue 08) - Scopus Indexed

- Proposed a novel hybrid cryptographic model combining ElGamal and Paillier encryption schemes to enhance security and computational efficiency in secure data processing.

ACHIEVEMENTS

- Scored 95.1% in ICSE and 91.2% in ISC, maintaining a top 5 rank in school throughout high school.
- Awarded Subject Topper with a perfect score of 100/100 in Social Science, ICSE 2021
- Recipient of Meritorious Award 2023 at event conducted by College Dekho and Dainik Jagran
- Ranked 1st in BCA for five consecutive semesters, demonstrating consistent academic excellence
- Participated in Inter College Coding Competition at ITS Ghaziabad (Samagra)
- Represented our college at ITS Ghaziabad (Samagra) Quiz Competition and secured 1st position
- Volunteered in organizing college-level technical fest
- Recipient of Arjuna Award 2025 for academic excellence and leadership at Udaan 4.0, Lucknow.
- Presented the research paper on Technology Enabled Promotion of Indian Knowledge Traditions in Contemporary Education in the CKNKH National Seminar 2026.